

ETSI FRMCS

Application Session Concept

Proposed Modifications to ETSI TS 103 764, TS 103 765-3/4/5

Working Contribution – Draft v2.9

R2DATO D25.3 – Concept Paper FRMCS

Target documents: ETSI TS 103 764 (RT-0095), TS 103 765-3 (RT-0097), TS 103 765-4 (RT-0099), TS 103 765-5 (RT-00104)

Unchanged documents: All UIC specifications (FFFIS-7950, FIS-7970, FRS FU-7120, AT-7800, TOBA FRS-7510, EIRENE FRS, EIRENE SRS)

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1 Architectural Principle

1.1 Reformulation of the Session Concept

In the ETSI TSs, “session” previously referred to the MCDATA IPcon tunnel established by the Gateway on behalf of an application (what this proposal calls a **service session**). This proposal redefines the terms:

- “**session**”: the application session – communication between two FRMCS railway applications, composed of one or more service sessions managed internally by the Gateways.
- “**service session**”: an individual communication bearer. The first is necessarily MCx (MC-Data IPcon) – unchanged. Additional ones may be any MCx type or future railway critical services (R2DATO D25.3 Part B); not limited to the same service domain.

All internal MCx tunnel occurrences of “session” in the ETSI TSs are replaced by “service session”. Application-visible identifiers (`sessionId`, `POST /sessions`, `openSession*`, `closeSession*`, `incomingSessionNotif*`, `open session`) are unchanged.

Nothing changes for applications. OB_{APP} and TS_{APP} interfaces are strictly unchanged. UIC specs already use “session” in the application sense – their terminology is now naturally aligned without modification.

1.2 Why ASAS Rather than SIP INVITE AnyExt

RT-0097 §6.2.2.3 already carries `<application-data>` in the `AnyExt` field (format “not specified in the present document”). While usable for MCx, this approach is structurally excluded: a future non-MCx critical service has no SIP INVITE to carry the negotiation. The protocol-agnostic ASAS is the only mechanism that survives the application/service strata separation of R2DATO D25.3 Part B.

1.3 Application Identifier Resolution (AIR)

The AIR is an internal Gateway function resolving `remoteId` to a **service identifier** (MCx or future critical service). First-edition rules – fully specified:

1. If `remoteId` ends with `.ertms` (ETCS format `id.ty.cc.ertms` as specified in UIC AT-7800 §5.3, which references UNISIG SUBSET-037): append the current MCx domain → `id.ty.cc.ertms@current`. Consistent with current UIC specifications; functional baseline (note: non-optimal in domain-overlap zones).
2. Otherwise: pass-through as MCx FA – identical to historical behaviour.

The AIR function is designed as an **extension point**. Future editions may introduce DNS-based or FROP-configured resolution without any change to OB_{APP} , TS_{APP} , SIP, or UIC specifications, in connection with ERJU work on ETCS addressing in FRMCS (UIC EIRENE SRS, UNISIG SUBSET-037).

1.4 Initialisation Mechanism

Once the first service session is established (SIP + GRE unchanged), each Gateway exposes an ASAS reachable on its post-NAT IP address (`SD_PUB_IP`, cf. §2.5). The calling Gateway drives:

1. **Initialisation**: POST to ASAS with `svcSessionId_1` as correlator; receives `appSessionId_term`, Ed25519 public key, alias list, MSP policy, keepalive period.

2. **Additional service sessions:** established per alias list + FROP; each declared to ASAS with signed `sessionToken`, enabling mutual authentication independently of the underlying signalling protocol.
3. **Keepalive:** sent only if applicative silence $\geq T_KA$ on a given `svcSessionRef` (intelligent keepalive, cf. §2.4).
4. **Alias list update:** PATCH sent immediately by terminating GW on any new alias or service identifier registration. No configuration required.
5. **Release:** DELETE to ASAS + release each service session per its signalling protocol.

1.5 Application Session Lifecycle

A session is alive while at least one service session is **ACTIVE** or **DEGRADED**. When all are lost, a **recovery timer** starts: INTEGER(1..180) s (FROP); SHALL be ≤ 5 s for sessions carrying ETCS/ATP (AT-7800 Annex A.6). On expiry, the **application session** is released.

2 Key Concepts

2.1 Terminology

Term	Definition
session	Application-level communication between two FRMCS railway applications, composed of one or more service sessions. Identified externally by <code>sessionId</code> (OB _{APP} /TS _{APP}), internally by (<code>appSessionId_call</code> , <code>appSessionId_term</code>).
service session	Individual communication bearer, transparent to applications. First is necessarily MCx (MCData IPcon). Additional may be MCx or future critical services. Identified by <code>svcSessionRef</code> within its parent session.
open session	A session that has not been released (unchanged).
AIR	Internal Gateway function resolving <code>remoteId</code> to a service identifier (MCx or future critical service). First-edition rules (fully specified): <code>*.ertms</code> $\rightarrow$ <code>*.ertms@current_domain</code> (UIC AT-7800 §5.3, ref. UNISIG SUBSET-037); otherwise pass-through FA MCx. Extension point for future optimisation (DNS/FROP, in connection with ERJU work on ETCS addressing – UIC EIRENE SRS, UNISIG SUBSET-037). Not visible to applications.
ASM	Internal Gateway function managing session and service session lifecycle, driving ASAS interactions and MSP policy. Not visible to applications.
ASAS	REST API server exposed by each Gateway on its post-NAT IP (SD_PUB_IP). Security is based on per-message Ed25519 authentication; there is no normative requirement for confidentiality of ASAS exchanges between Gateways. HTTP transport MAY be protected by TLS (recommended). Ed25519 key exchange anchored on first MCx service session – no inter-GW PKI required.

Term	Definition
MSP	Multi Service Path: inter-Gateway mechanism managing multiple parallel service sessions within a single application session, operated by the ASMs. Distinct from transport-level Multipath (RT-00104). Policy (<code>mSPPolicy</code>) negotiated at ASAS initialisation: DUPLICATION, RECOVERY or LOADBALANCE.
Calling GW	GW initiating <code>POST /sessions</code> via <code>OB_{APP}</code> . Derives ASAS IP from post-NAT SIP source IP.
Terminating GW	GW hosting destination application. Exposes ASAS. Builds alias list; SHALL immediately PATCH on any new alias or service identifier registration; notifies application via <code>TS_{APP} (incomingSessionNotif*)</code> .
alias list	Dynamic list of service identifiers + types + domains. Returned at initialisation; immediately updated via <code>PATCH /aliases</code> on any new registration – no FROP configuration required.
service identifier	Identifier used to establish a service session, regardless of service type. For MCx: FA or MC Service User ID. For future critical services: any identifier the terminating Gateway can associate with the destination user.
recovery timer	INTEGER(1..180)s (FROP); SHALL be ≤ 5 s for ETCS/ATP (AT-7800 Annex A.6). Application session released on expiry.
T_KA	Keepalive period: INTEGER(1..10)s (application profile). Rule: $T_KA \leq \text{recovery_timer} / 2$ (≤ 2.5 s for ETCS/ATP). $N = 2$ consecutive failures (fixed) $\rightarrow$ DEGRADED.

2.2 Session Identifiers

- `appSessionId_call` (UUID v4): calling GW, stable for session lifetime.
- `appSessionId_term` (UUID v4): terminating GW at ASAS init. Analogous to GTP TEIDs.
- `svcSessionRef` (UUID v4): identifies a service session. **Stable and immutable**, including during recovery.
- `svcSessionId` (native): SIP Call-ID for MCx. Used as correlator. May change during recovery.

2.3 Authentication Model

The first MCx service session is a trusted service already responsible for identifying the parties – its establishment is a sufficient authentication anchor. There is no normative requirement for confidentiality of ASAS exchanges between Gateways: the ASAS operates in a railway inter-GW environment where protection of session metadata is not a normative prerequisite.

Level 1 – HTTP Transport (TLS recommended)

The ASAS HTTP transport MAY be protected by TLS 1.3 (recommended for transport integrity). Server certificate may be **self-signed** – no inter-operator PKI required. TLS is not normative at this level; normative authentication is provided by Levels 2 and 3. The mTLS option is possible if a bilateral OOB FSNNI agreement so provides (FROP).

FSNNI topology hiding note: a potential intermediate domain performing topology hiding does not compromise authentication – the Ed25519 `sessionToken` provides per-message integrity regardless of the network path. The only constraint is ASAS reachability on SD_PUB_IP (covered by §2.5).

Level 2 – Authentication Anchor (First MCx Service Session)

The MC Service Server authenticated the calling Gateway before accepting the session. Ed25519 public key exchange over this channel bootstraps authentication: any key received at ASAS initialisation is authentic by construction.

Level 3 – Per-Message Authentication (Ed25519 sessionToken)

```
sessionToken = Sign(sk_call,
  appSessionId_call || 0x00 || appSessionId_term || 0x00 ||
  svcSessionRef      || 0x00 || svcSessRequestTime || 0x00 ||
  svcSessAcceptTime)
// Recovery: append 0x00 || replSvcSessionRef
ackToken = Sign(sk_term, same input)
```

Serialisation: UTF-8 concatenation with 0x00 separators – deterministic, no external dependency.

2.4 Intelligent Keepalive

- T_KA: INTEGER(1..10) s (application profile); SHALL be ≤ 2.5 s for ETCS/ATP.
- $N = 2$ consecutive failures (fixed) $\rightarrow$ DEGRADED.
- Recovery: new service session declared with same `svcSessionRef` + signed token.

2.5 NAT Addressing Model

ASAS address rule (resolved): post-NAT source IP of first service session SIP signalling (SD_PUB_IP, RT-0097 §6.2.2.4). No SIP header extraction required. Port configurable in FROP.

3 Terminology Replacement Rule in ETSI TSs

Context in TSs	Current term	Proposed term
Internal MCx bearer	session	service session
Internal MCx establishment	session establishment	service session establishment
Internal MCx release	session release	service session release
Inter-GW multi-path (service session level)	Multipath session	Multi Service Path (MSP)
Transport-level Multipath (RT-00104)	Multipath	Multipath (unchanged)
OB _{APP} /TS _{APP} identifier	sessionId	unchanged
Application request	POST /sessions, openSession*	unchanged
Incoming notification	incomingSessionNotif*	unchanged

Context in TSs	Current term	Proposed term
Application release	<code>closeSession*</code> , <code>DELETE /sessions</code>	unchanged
Application state	<code>open session</code>	unchanged

4 Compatibility with single-UE IFDT (RT260101021)

4.1 Context

UIC contribution RT260101021 examines three approaches for Inter-FRMCS Domain Transition (IFDT) in a single-UE context (a single onboard RM), applicable to LC applications (ETCS/ATO). It proposes three options — A (dual MC ID), B (migration of a single MC ID), C (parallel sessions with LBO/HR) — and introduces the notion of `sessionId` as a possible enabler for service continuity during transitions. The present proposal is orthogonal to IFDT: it operates at the application session level between GWs, independently of underlying 5G mobility events.

4.2 Analysis per IFDT option

Option A — Dual MC ID (make-before-break) The EVC maintains MC ID1 in the serving FSD and establishes MC ID2 in the target FSD in parallel. Each MC ID corresponds to a distinct service session at the OB GW level. Compatibility: native. The MSP mechanism is precisely designed for this scenario: both service sessions (MC ID1 towards the serving FSD, MC ID2 towards the target FSD) coexist within the same application session. The MSP RECOVERY policy allows the ASM to switch the application data plane from the active service session to the backup service session as soon as it becomes available, without any visible interruption for the EVC. The `svcSessionRef` of the MC ID2 service session is declared to the ASAS upon establishment (§x.4 - 5.1), which precisely implements the “`sessionId` concept for recovery” identified in RT260101021.

Option B — Single MC ID migration After PLMN reselection, MC ID1 migrates to the target FSD (MC ID1a); there is an interruption during migration followed by re-establishment. Compatibility: full, via the recovery mechanism. The loss of the MC ID1 service session (during migration) triggers the recovery timer on the OB ASM side. As soon as MC ID1a is registered in the target FSD, the OB GW establishes a new service session towards MC ID1a and declares it to the ASAS with the same `svcSessionRef` + `replSvcSessionRef` (recovery token, §x.4 - 5.1). The EVC application session is resumed without observable interruption, provided that the total migration duration remains within the recovery timer (≤ 5 s for ETCS). The interruption inherent to Option B is therefore absorbed by the recovery timer — which is consistent with constraint AT-7800 Annex A.6.

Option C — Parallel LBO + HR sessions (SSC mode 1) An LBO path (via the target FSD) and an HR path (preserved via the serving FSD) coexist; N14/N38 and cross-domain N5/Rx are required at the 5GC level. Compatibility: full, with added value. The LBO and HR paths correspond exactly to two parallel service sessions within the application session, managed by MSP in DUPLICATION mode. The ASAS provides the application correlator (`appSessionId_call/appSessionId_term`) which links the two service sessions, which RT260101021 identifies as a requirement for “multi-path switching logic”. The cross-domain complexity (N14, N38, N5/Rx) remains entirely within the 5GC and is not exposed to the ASAS. The decision to switch between LBO and HR remains local to the OB ASM, driven by ACTIVE/DEGRADED statuses returned by the ASAS keepalive (§x.5 - 5.1). Note on Options A and C: both options provide uninterrupted

application continuity for the EVC, but through different MSP mechanisms. Option A uses MSP RECOVERY mode (sequential switchover: the second service session is pre-declared but the data plane switches only when the first is lost) and remains constrained by the recovery timer (≤ 5 s for ETCS/ATP). Option C uses MSP DUPLICATION mode (native parallelism: both service sessions are active simultaneously) and is the most directly aligned with the concept of an application session: both service sessions naturally coexist within the session without any recovery timer, and switching is instantaneous. Option C is therefore the more robust of the two for ETCS, at the cost of higher 5GC complexity (N14, N38, cross-domain N5/Rx).

4.3 Contribution to single-UE formalisation in ETSI TS

RT260101021 asks TC RT to analyse single-UE options and consider their formalisation in ETSI TS before RT#102. The present proposal provides precisely the missing building blocks identified in that contribution: The sessionId concept for recovery/transfer (mentioned in RT260101021) corresponds to the pair (appSessionId_call, appSessionId_term) + immutable svcSessionRef of the present proposal. The GRE tunnel switching mentioned in RT260101021 is covered by the ASAS recovery mechanism (§X.4 - 5.1, replSvcSessionRef), which allows a new GRE bearer to be attached to the existing application session. The multi-path switching logic (Option C) is covered by the MSP DUPLICATION policy negotiated at ASAS initialisation. The present proposal can therefore serve as a common normative basis for all three single-UE IFDT options, avoiding duplication of correlation and recovery logic in each option. It is recommended to explicitly highlight this in the ETSI response to RT260101021.

5 Proposed ETSI Modifications

5.1 ETSI TS 103 764 (RT-0095) – Normative Annex X

X.1 – Scope ASAS inter-GW API, agnostic to service session type. Not visible to applications. Does not modify any UIC specification or service protocol. Absorbs FUTURE_CRITICAL_SERVICE without modification of this annex.

X.2 – Transport and Authentication Transport: HTTP/2 + JSON. Port: FROP. HTTP transport MAY be protected by TLS 1.3 (recommended); there is no normative requirement for confidentiality of ASAS exchanges between Gateways.

Server certificate (if TLS): Self-signed or any locally-deployed CA – no inter-operator PKI required.

ASAS IP address: Post-NAT source IP of first service session SIP signalling (SD_PUB_IP, RT-0097 §6.2.2.4). No SIP header extraction required.

Normative authentication: First MCx service session anchor + Ed25519 keys exchanged over that trusted channel. All subsequent ASAS exchanges authenticated by Ed25519 signature. No inter-GW PKI required.

Token serialisation: UTF-8 fields concatenated with 0x00 separators (see §2.3).

Timers: recovery timer INTEGER(1..180)s (FROP); T_KA INTEGER(1..10)s (application profile); $N = 2$ (fixed).

mTLS option: If bilateral OOB FSNNI agreement (FROP).

X.3 – Initialisation (POST /asas/appsessions)

```
// Request
{ "appSessionId_call": "uuid-v4", "svcSessionRef_1": "uuid-v4",
```



```
"svcSessionId_1": "native-id", "callingPublicKey": "base64(Ed25519)",
"keepaliveReq": 2,
"mspCapabilities": { "supportedPolicies": ["DUPLICATION","RECOVERY","
LOADBALANCE"],
                    "maxPaths": 4 } }
```

```
// Response 200 OK
{ "appSessionId_term": "uuid-v4", "svcSessionRef_1": "uuid-v4",
  "terminatingPublicKey": "base64(Ed25519)", "keepaliveAccepted": 2,
  "mspPolicy": { "policy": "DUPLICATION|RECOVERY|LOADBALANCE|NONE", "
    activePaths": 2 },
  "advertisedAliases": [{ "svcSessionRef": "uuid-v4",
    "serviceType": "MCX|FUTURE_CRITICAL_SERVICE",
    "serviceDomain": "string", "serviceId": "string",
    "reachability": "CURRENT|ALTERNATIVE" }] }
```

X.4 – Service Session Declaration / Recovery

```
// Request
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "svcSessionRef": "uuid-v4", "serviceType": "MCX|FUTURE_CRITICAL_SERVICE",
  "svcSessionId": "native-id", "svcSessRequestTime": "ISO8601",
  "svcSessAcceptTime": "ISO8601",
  "sessionToken": "base64(Sign(sk_call, quintet_0x00_separated))" }
// Recovery: add "replSvcSessionRef": "uuid-v4"

// Response 200 OK
{ "svcSessionRef": "uuid-v4",
  "ackToken": "base64(Sign(sk_term, same input))" }
```

X.5 – Keepalive Sent only if applicative silence $\geq T_{KA}$ on listed svcSessionRef.

```
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "svcSessionRefs": ["uuid-v4"], "timestamp": "ISO8601" }
// Response: { "results": [{ "svcSessionRef": "uuid-v4", "status": "
ACTIVE|DEGRADED" }] }
```

X.6 – Alias List Update (PATCH .../aliases) Sent immediately by terminating GW on any new alias or service identifier registration during an active session. No FROP configuration required.

```
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "updatedAliases": [{ "svcSessionRef": "uuid-v4",
    "serviceType": "MCX|FUTURE_CRITICAL_SERVICE",
    "serviceDomain": "string", "serviceId": "string",
    "reachability": "CURRENT|ALTERNATIVE", "action": "ADD|REMOVE" }] }
```

X.7 – Release (DELETE .../appsessions/{id})

```
{ "appSessionId_call": "uuid-v4", "appSessionId_term": "uuid-v4",
  "reason": "NORMAL|ERROR|TIMEOUT" }
```

X.8 – State Machines Session:

[CREATING]	-- 200 OK	--> [ESTABLISHED]
[CREATING]	-- 4xx/5xx	--> [FAILED]
[ESTABLISHED]	-- all svc lost	--> [DEGRADED]

```
[ESTABLISHED] -- DELETE /sessions --> [RELEASING]
[DEGRADED]    -- >=1 ACTIVE      --> [ESTABLISHED]
[DEGRADED]    -- timer expired   --> [RELEASING]
[RELEASING]   -- done            --> [RELEASED]
```

Service session:

```
[CREATING]      -- bearer+ASAS OK      --> [ACTIVE]
[ACTIVE]        -- 2 KA no response    --> [DEGRADED]
[DEGRADED]      -- recovery            --> [ACTIVE]
[ACTIVE|DEGRADED] -- RELEASING          --> [RELEASED]
```

X.9 – Error Mapping to OB_{APP} Causes

ASAS cause	openSessionFinalAnswerNotif cause
ASAS timeout	MCX_ENDPOINT_NOT_REACHABLE
ASAS 403	REMOTE_ENDPOINT_DECLINED
ASAS 404	TERMINATING_APPLICATION_ENDPOINT_NOT_REACHABLE
ASAS 401	TERMINATING_APPLICATION_ENDPOINT_NOT_ALLOWED

6 Impact Summary

Document	Nature of modification	UIC specs
ETSI TS 103 764	Session + service session + AIR + MSP + ASAS + service identifier (§3.1) + ASM + Annex X	
ETSI TS 103 765-3	Systematic replacement + Step 1 (AIR → service id + SD_PUB_IP) + MSP policy	unchanged
ETSI TS 103 765-4	Systematic replacement + Step 0 + immediate alias list mgmt + MSP	unchanged
ETSI TS 103 765-5	Systematic replacement + MSP binding + Multipath/MSP distinction	unchanged
UIC FFFIS-7950	No modification	✓
UIC FIS-7970	No modification	✓
UIC FRS FU-7120	No modification	✓
UIC AT-7800 SRS	No modification	✓
UIC TOBA FRS-7510	No modification	✓
3GPP TS 24.582/24.379	No modification	

7 Open Points

All open points are resolved. The document is ready for ETSI contribution.

1. ASAS IP / NAT – Resolved (§2.5).
2. FSNNI multi-hop trust – Resolved (§2.3).
3. sessionToken serialisation – Resolved (§2.3 + §X.2).
4. AIR rules – Resolved: *.ertms → @current_domain; otherwise pass-through.

- 5. Recovery timer – Resolved: $\text{INTEGER}(1..180)\text{ s}; \leq 5\text{ s}$ for ETCS/ATP.
- 6. T_{KA} / N – Resolved: $\text{INTEGER}(1..10)\text{ s}; N = 2; T_{\text{KA}} \leq \text{timer}/2$.
- 7. Alias list triggering – Resolved: immediate, no FROP.
- 8. Extensibility – Resolved: `serviceType` open field.